

# **TITLE PAGE**

- **Problem Statement ID – 1**
- **Problem Statement Title- “Eco-Loop Building Agents”**
- **Theme- Smart Automation & Sustainable Development.**
- **PS Category- Software.**
- **Student Name (Registered on portal)- Manikya Nariyal**
- **Student ID- 23BCE11009**

# IDEA TITLE

## **IDEA TITLE Eco-Loop: AI-Powered Closed-Loop Building Energy Optimization**

### **Proposed Solution**

- Closed-loop system combining EnergyPlus simulation with local LLM (Qwen2.5) .
- AI analyzes results, recommends HVAC setpoint, auto-modifies IDF and re-simulates.
- Generates interactive dashboard, PDF/CSV reports and tracks optimization history.

### **How it addresses the problem**

- Removes manual analysis of large EnergyPlus outputs and repeated IDF editing .
- Automates full optimize-verify cycle and delivers measurable electricity savings.
- Reduces dependency on domain experts for continuous HVAC schedule tuning.

### **Innovation & Uniqueness**

- Fully local LLM with no cloud dependency and true closed-loop verification.
- Automatic IDF modification plus interactive dashboard and professional reports.

# TECHNICAL APPROACH

## Technologies Used

- Python 3.13, EnergyPlus 26.1, Ollama + Qwen2.5 1.5B (local LLM) .
- Streamlit, Plotly, Pandas, ReportLab for dashboard and automated reports .
- IDF building models and EPW weather files for accurate simulation .
- Modular architecture supporting easy extension to new optimization strategies.

## Methodology & Implementation Process

- Execute baseline EnergyPlus simulation and extract key energy metrics
- Feed metrics to local LLM to generate optimal HVAC cooling setpoint .
- Auto-update IDF, re-simulate and calculate energy savings
- Produce interactive dashboard, PDF/CSV/JSON reports and maintain run history.

# FEASIBILITY AND VIABILITY

## Feasibility of the Idea

- Uses existing open-source tools (EnergyPlus, Ollama, Streamlit).
- Fully functional prototype already developed and tested with real savings (3.57%)
- Runs completely offline on standard hardware with local LLM

## Potential Challenges & Risks

- Limited accuracy of small local LLM for complex multi-zone buildings
- EnergyPlus simulation time increases with larger building models
- Dependence on quality of weather data and initial IDF model

## Strategies to Overcome Challenges

- Upgrade to larger local models or fine-tune Qwen for HVAC domain
- Use parallel simulation and caching for faster iterative runs
- Add validation checks and fallback default setpoints for robustness

| Metric                   | Baseline        | Optimized       |
|--------------------------|-----------------|-----------------|
| Outdoor Temp (°C)        | 25.58           | 25.58           |
| Building Electricity (J) | 257,872,140,000 | 257,872,140,000 |
| CO2                      | 7,653,423,040   | 364,161,184,779 |

A row of three theme selection icons. The first icon is a circle with a vertical line through the center, labeled 'System'. The second icon is a sun with rays, labeled 'Light'. The third icon is a crescent moon, labeled 'Dark'.

- Rerun R
- Auto rerun ☒
- Clear cache C
- Print
- Record screen
- Made with Streamlit v1.60.0

| Parameter            | Baseline            |
|----------------------|---------------------|
| Facility Electricity | 377,653,423,040 kWh |
| Building Electricity | 22,872,342,000 kWh  |
| Outdoor Temperature  | 25.8°C              |
| Cooling Setpoint     | 23.9°C              |

The diagram illustrates the Optimization Pipeline, which consists of two sequential steps:

- 01 Simulation**: EnergyPlus baseline execution
- 02 Analysis**: AI performance evaluation

### AI Recommendation Summary

| RECOMMENDED SETPOINT | EXPECTED SAVINGS |
|----------------------|------------------|
| 25.0°C               | 3.57%            |

**DECISION ANALYSIS**

The AI model analyzed baseline performance and selected an optimal cooling setpoint to reduce HVAC energy consumption while maintaining thermal comfort standards.

# RESEARCH AND REFERENCES

- EnergyPlus Official Documentation – <https://energyplus.net>
- Ollama & Qwen2.5 Model Card – <https://ollama.com> & <https://huggingface.co/Qwen>
- ASHRAE Guidelines for HVAC Setpoint Optimization
- Building Energy Simulation Best Practices (IBPSA & DOE Resources)
- Streamlit & Plotly Documentation for Interactive Dashboards
- ReportLab Documentation for Automated PDF Report Generation

## **Additional Supporting Work**

- Research papers on AI-assisted building energy management and model-predictive control
- Open-source EnergyPlus IDF examples and weather data (EPW) from EnergyPlus Weather Database